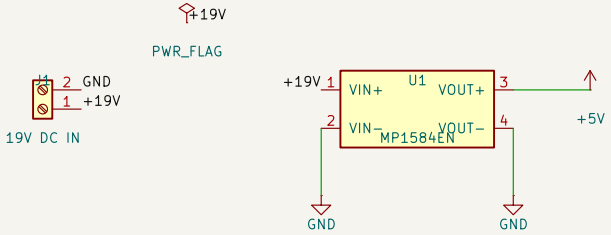


DESIGN NOTES

1. SET U1 OUTPUT TO 5.0 V BEFORE PLUGGING IN U2. The MP1584 output is set by a trimpot; the as−shipped setting is arbitrary and will destroy the D1 mini if it is above 5 V.

2. R1 IS THE ACCURACY−CRITICAL PART: 100R 0.1%, <=50 ppm/K metal film, >=0.25 W. Carbon drifts up to 1000 ppm/K and the shed sees −10 to +40 C. Do not substitute.

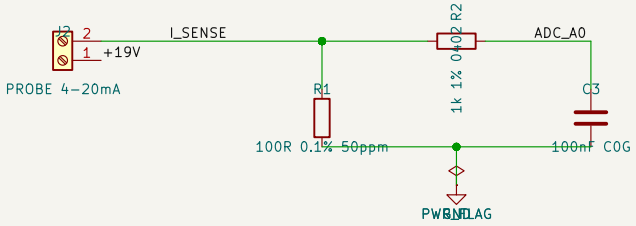
3. GROUND ROUTING: take R1's low side directly to U3's GND pin, not through the ESP's return path. 200 mA of WiFi burst through shared ground = ~2 mV = 0.12 % of full scale.

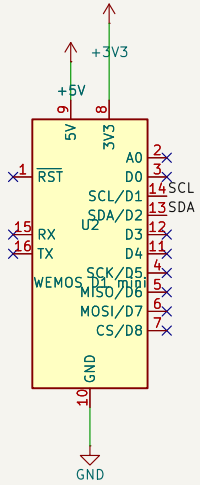
4. Burden: 4 mA = 0.400 V, 20 mA = 2.000 V, cable break = 0.000 V. The 0 V reading is what separates a broken cable from an empty tank. Calibrate two−point across R1.

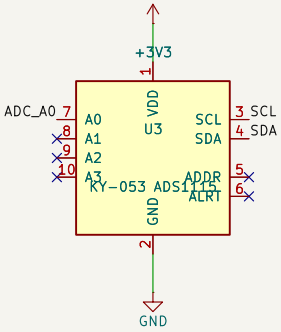
5. NO INPUT PROTECTION FITTED — no fuse, no reverse−polarity diode, no TVS. Feed only from a known−good 19 V PSU, and check J1 polarity before every power−up.

6. J2 POLARITY MUST BE SILKSCREENED. A reversed probe reads 0 mA — indistinguishable from a cable break. Probe needs <17 V compliance (19 V minus the 2 V burden drop).

7. OPEN ITEMS: verify U1 (MP1584) and U3 (KY−053) pad pitch and pin order before ordering.







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